Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – IV

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Part A - Theory:

1. What is ASP.NET Web Forms?
ASP.NET Web Forms is a web application framework developed by Microsoft for building dynamic, interactive, and data-driven websites. It provides a rich collection of **server controls** that simplify web application development by reducing the amount of code required.
Features of ASP.NET Web Forms:

- Event-driven programming model.
- Rich collection of server controls.
- Built-in state management using ViewState.
- Easy integration with databases.
- Support for validation controls.
- Rapid application development.

2. What are Server Controls?
Server Controls are controls that run on the server and generate HTML dynamically. They simplify user interaction and automatically handle client-server communication.
Syntax:
<asp:ControlName ID="ControlID" runat="server" />

3. Common ASP.NET Server Controls
a) Label Control
The **Label** control is used to display static text or messages on a web page.
Syntax
<asp:Label ID="lblName" runat="server" Text="Student Name"></asp:Label>

b) TextBox Control
The **TextBox** control allows the user to enter text.
Syntax
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>

c) RadioButton Control
The **RadioButton** control allows the user to select only one option from a group.
Syntax

```
<asp:RadioButton ID="rbMale" runat="server" Text="Male" GroupName="Gender" />
```

d) DropDownList Control
The **DropDownList** control displays a list of options from which the user can select one.
Syntax

```
<asp:DropDownList ID="ddlCourse" runat="server">
  <asp:ListItem>BSc IT</asp:ListItem>
  <asp:ListItem>BSc CS</asp:ListItem>
  <asp:ListItem>BCA</asp:ListItem>
</asp:DropDownList>
```

e) CheckBox Control
The **CheckBox** control allows the user to select or deselect an option.
Syntax

```
<asp:CheckBox ID="chkTerms" runat="server" Text="I Agree" />
```

f) Calendar Control
The **Calendar** control allows users to select a date from a graphical calendar.
Syntax

```
<asp:Calendar ID="Calendar1" runat="server"></asp:Calendar>
```

g) Button Control
The **Button** control is used to perform an action when clicked.
Syntax

```
<asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
```

4. Validation Controls
Validation controls are used to ensure that the user enters valid data before submitting a form.
Types of Validation Controls
a) RequiredFieldValidator
Ensures that a control is not left empty.

```
<asp:RequiredFieldValidator
ID="rfvName"
runat="server"
ControlToValidate="txtName"
ErrorMessage="Name is required">
</asp:RequiredFieldValidator>
```

b) RangeValidator
Checks whether a value lies within a specified range.

c) CompareValidator
Compares the value of one control with another control or a fixed value.

d) RegularExpressionValidator
Validates the input using a regular expression (e.g., Email, Mobile Number).

e) CustomValidator
Allows custom validation logic written by the programmer.

f) ValidationSummary
Displays all validation error messages in one place.

5. AdRotator Control
The **AdRotator Control** displays different advertisements or images randomly from an XML file. It is commonly used to display advertisements, announcements, offers, or promotional banners.
Advantages

- Displays advertisements automatically.
- Randomly rotates images.
- Reads advertisement details from an XML file.
- Easy to maintain and update.

Syntax
```
<asp:AdRotator
ID="AdRotator1"
runat="server"
AdvertisementFile="Ads.xml">
</asp:AdRotator>
```

6. Advantages of ASP.NET Server Controls

- Easy to use and configure.
- Supports event-driven programming.
- Reduces coding effort.
- Provides automatic state management.
- Supports built-in validation.
- Improves application security.
- Generates browser-compatible HTML automatically.

7. Applications

- Student Registration System
- Online Admission Portal
- Employee Registration System

- Online Examination System
- Library Management System
- E-Commerce Registration Forms
- Hospital Appointment System

Part B - Program

a) Create an ASP.NET Web Forms application using simple server controls.

a. Design a Student Registration Form using the following ASP.NET controls:

i. Label and TextBox to accept Student Name
ii. RadioButton to select Gender (Male/Female)
iii. DropDownList to select Course (BSc IT, BSc CS, BCA)
iv. CheckBox for accepting Terms & Conditions
v. Calendar control to select Date
vi. Button control to submit the form
vii. Label control to display the output

Algorithm:

Step 1: Start the application.
Step 2: Display the form with Name, Gender, Course, Terms & Conditions, Calendar, and Submit button.
Step 3: Accept the student's name in the TextBox.
Step 4: Select the student's gender using the RadioButtonList.
Step 5: Select the course from the DropDownList.
Step 6: Accept the Terms & Conditions using the CheckBox.
Step 7: Select a date from the Calendar.
Step 8: Click the **Submit** button.
Step 9: Read the entered Name, selected Gender, Course, and Date.
Step 10: Display the entered details in the Label.
Step 11: Stop the application.

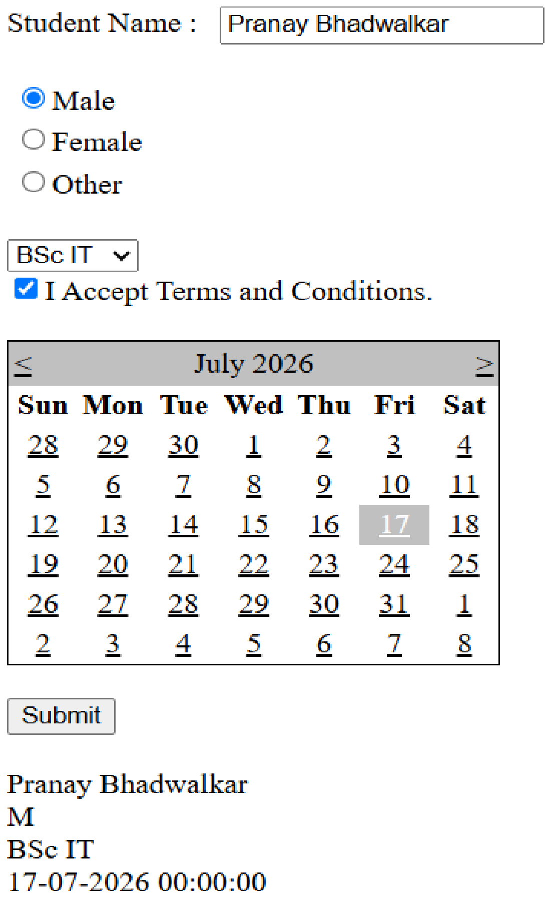
Code:
Default.aspx.cs:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Default3.aspx.cs" Inherits="Default3" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:Label ID="Label1" runat="server" Text="Student Name :"></asp:Label>
         
        <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>

        <br /><br />

        <asp:RadioButtonList ID="RadioButtonList1" runat="server">
            <asp:ListItem Value="M">Male</asp:ListItem>
            <asp:ListItem Value="F">Female</asp:ListItem>
            <asp:ListItem Value="O">Other</asp:ListItem>
        </asp:RadioButtonList>

        <br />

        <asp:DropDownList ID="DropDownList1" runat="server">
            <asp:ListItem>BSc IT</asp:ListItem>
            <asp:ListItem>BSc CS</asp:ListItem>
            <asp:ListItem>BCA</asp:ListItem>
        </asp:DropDownList>

        <br />

        <asp:CheckBox ID="CheckBox1" runat="server" Text="I Accept Terms and Conditions." />

        <br /><br />
```

Default.aspx:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

2 references
public partial class Default3 : System.Web.UI.Page
{
    0 references
    protected void Page_Load(object sender, EventArgs e)
    {

    }

    0 references
    protected void Button1_Click(object sender, EventArgs e)
    {
        string name = TextBox1.Text;
        string gender = RadioButtonList1.Text;
        string course = DropDownList1.Text;
        string date = Calendar1.SelectedDate.ToString();

        Label2.Text = name + "<br/>" + gender + "<br/>" + course + "<br/>" + date;
    }
}
```

Output:



b) Create a Registration form to demonstrate use of various Validation controls.

Algorithm:

Step 1: Start the application.
Step 2: Display the form with Name, Age, Password, and Confirm Password fields.
Step 3: Accept the user input.
Step 4: Check whether the Name field is empty.
Step 5: Validate that the Age is between 18 and 99.
Step 6: Compare the Password and Confirm Password fields.
Step 7: If any validation fails, display the appropriate error message.
Step 8: If all validations are successful, submit the form.
Step 9: Stop the application.

Code:

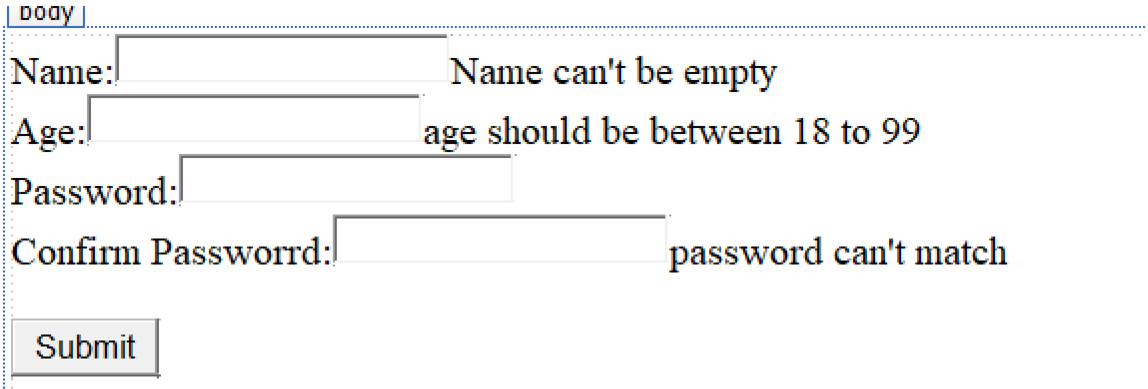
Client Objects & Events | (No Events)

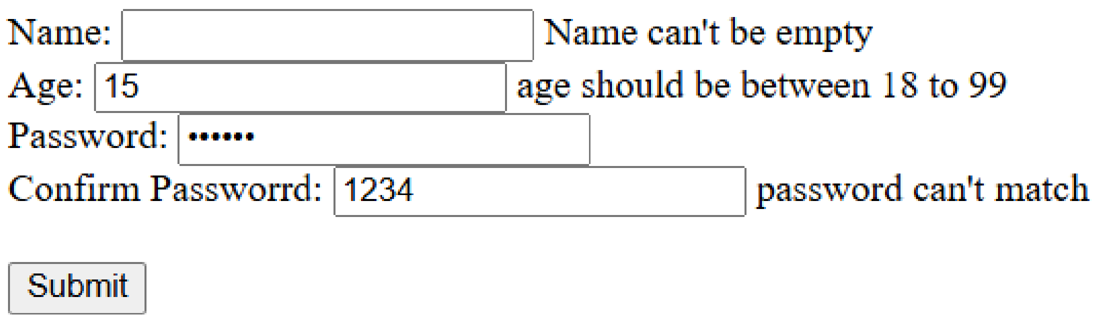
```
</head>
<body>
    <form id="form1" runat="server">
        <div>

            <asp:Label ID="Label1" runat="server" Text="Name:"></asp:Label>
            <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
            <asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server" ControlToValidate="TextBox1" ErrorMessage="Name can't be empty"></asp:RequiredFieldValidator>
            <br />
            <asp:Label ID="Label2" runat="server" Text="Age:"></asp:Label>
            <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
            <asp:RangeValidator ID="RangeValidator1" runat="server" ControlToValidate="TextBox2" ErrorMessage="age should be between 18 to 99" MaximumValue="99" MinimumValue="18"></asp:Range
            <br />
            <asp:Label ID="Label3" runat="server" Text="Password:"></asp:Label>
            <asp:TextBox ID="TextBox3" runat="server" TextMode="Password"></asp:TextBox>
            <br />
            <asp:Label ID="Label4" runat="server" Text="Confirm Passwordd:"></asp:Label>
            <asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
            <asp:CompareValidator ID="CompareValidator1" runat="server" ControlToCompare="TextBox3" ControlToValidate="TextBox4" ErrorMessage="password can't match"></asp:CompareValidator>
            <br />
            <br />
            <asp:Button ID="Button1" runat="server" Text="Submit" />

        </div>
    </form>
</body>
</html>
```

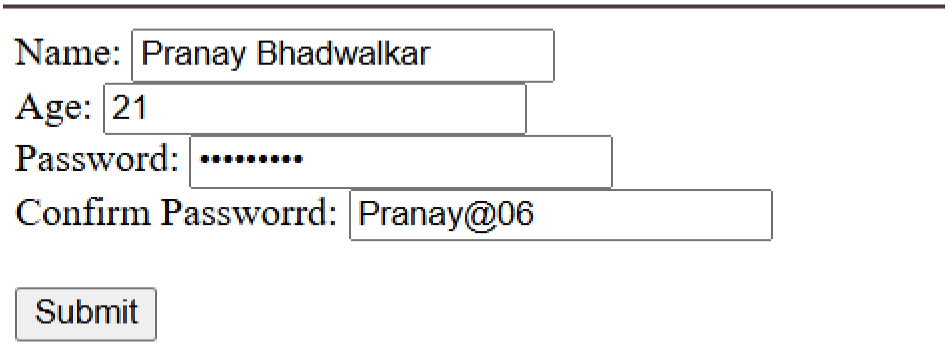
91 % | No issues found | Ln: 35, Ch: 1 | SPC | CRLF

Design | Split | Source

Output:






Name: Pranay Bhadwalkar
Age: 21
Password: ••••••••••
Confirm Passworrd: Pranay@06
Submit

c) Create Web Form to demonstrate use of Adrotator Control.

Algorithm:
Step 1: Start the application.
Step 2: Display the form with Name, Age, Password, Confirm Password, Submit button, and AdRotator.
Step 3: Accept the user's Name.
Step 4: Validate that the Name field is not empty using RequiredFieldValidator.
Step 5: Accept the user's Age.
Step 6: Validate that the Age is between 18 and 99 using RangeValidator.
Step 7: Accept the Password and Confirm Password.
Step 8: Compare both passwords using CompareValidator.
Step 9: If any validation fails, display the corresponding error message.
Step 10: If all validations are successful, submit the form.
Step 11: Read the advertisement details from XMLFile.xml.
Step 12: Display the advertisement using the AdRotator control.
Step 13: Stop the application.

Code:
Default2.aspx:

XMLFile.xml | Default2.aspx | Default.aspx

Client Objects & Events | (No Events)

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Default2.aspx.cs" Inherits="Default2" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>

            <asp:Label ID="Label1" runat="server" Text="Name:"></asp:Label>
            <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
            <asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server" ControlToValidate="TextBox1" ErrorMessage="Name can't be empty"></asp:RequiredFieldValidator>
            <br />
            <asp:Label ID="Label2" runat="server" Text="Age:"></asp:Label>
            <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
            <asp:RangeValidator ID="RangeValidator1" runat="server" ControlToValidate="TextBox2" ErrorMessage="age should be between 18 to 99" MaximumValue="99" MinimumValue="18"></asp:RangeValidator>
            <br />
            <asp:Label ID="Label3" runat="server" Text="Password:"></asp:Label>
            <asp:TextBox ID="TextBox3" runat="server" TextMode="Password"></asp:TextBox>
            <br />
            <asp:Label ID="Label4" runat="server" Text="Confirm Passworrd:"></asp:Label>
            <asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
            <asp:CompareValidator ID="CompareValidator1" runat="server" ControlToCompare="TextBox3" ControlToValidate="TextBox4" ErrorMessage="password can't match"></asp:CompareValidator>
            <br />
            <br />
            <asp:Button ID="Button1" runat="server" Text="Submit" />
            <br />
            <br />
            <asp:AdRotator ID="AdRotator1" runat="server" AdvertisementFile="~/XMLFile.xml" />

        </div>
    </form>
</body>
</html>
```

XMLFile.xml:

XMLFile.xml | Default2.aspx | Default.aspx

```
<?xml version="1.0" encoding="utf-8" ?>

<Advertisements>
    <Ad>
        <ImageUrl>Images/image1.png</ImageUrl>
        <NavigateUrl>https://www.google.com</NavigateUrl>
        <AlternateText>Collage Logo</AlternateText>
        <Height>100</Height>
        <Width>100</Width>
    </Ad>
    <Ad>
        <ImageUrl>Images/image2.png</ImageUrl>
        <NavigateUrl>https://www.google.com</NavigateUrl>
        <AlternateText>Collage Logo</AlternateText>
        <Height>100</Height>
        <Width>100</Width>
    </Ad>
</Advertisements>
```

Output:

XMLFile.xml | Default2.aspx | Default.aspx

form#form1

Name: Name can't be empty

Age: age should be between 18 to 99

Password:

Confirm Passworrd: password can't match

Submit

Name: Pranay Bhadwalkar

Age: 21

Password: ••••••

Confirm Passworrd: 123456

Submit

Name: Pranay Bhadwalkar

Age: 21

Password: ••••••

Confirm Passworrd: 123456

Submit